

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: B.Tech.

Branch : Mechanical Engineering

Semester :VI

Subject Code & Name: BTMEC605A_Y19 Quantitative Techniques in Project Management

Max Marks: 60

Date: 10/07/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q.1	Objective type questions. (Compulsory Question)		12
1	Operations Research approach is _____.	1	1
	a. multi-disciplinary b. scientific c. intuitive d. collect essential data		
2	A tie for leaving variable in simplex procedure implies	1	1
	a. optimality b. cycling c. No solution d. degeneracy		
3	In marking assignments, which of the following should be preferred?	2	1
	a. Only row having single zero b. Only column having single zero c. Only row/column having single zero d. Column having more than one zero		
4	The allocation cells in the transportation table will be called _____ cell.	2	1
	a. occupied b. unoccupied c. no d. finite		
5	The server utilization factor is also known as _____	3	1
	a. erlang distribution b. poisson distribution c. exponential distribution d. traffic intensity		
6	The problem of replacement is felt when job performing units fail _____.	3	1
	a. suddenly and gradually b. gradually c. suddenly d. neither gradually nor suddenly		
7	Which of the following is not an inventory?	4	1
	a. machine b. raw material c. finished goods d. consumable tools		
8	Larger size of inventory is a sign of -----	4	1
	a. better planning, b. inefficiency c. better scheduling d. none		
9.	In PERT, the distribution of activities times is assumed to be	5	1
	a. normal b. gamma c. beta d. exponential		
10	_____ is used for non-repetitive jobs	5	1
	a. Queue b. Replacement c. CPM d. PERT		

11	A activity in a network diagram is said to be _____ if the delay in its start will further delay the project completion time.				6	1																																										
	a. forward pass	b. backward pass	c. critical.	d. non-critical.																																												
12	What is the process of incorporating changes and rescheduling or replanning called?				6	1																																										
	a. Change of critical path	b. Decrease of project completion time	c. Increase of project completion time	d. All of the above																																												
Q. 2	Solve the following.					12																																										
A)	Define the following with respect to a linear programming problem 1. Feasible Solution 2. Optimal Solution 3. Corner point feasible solution				1	6																																										
B)	Solve the following LPP by graphical method Maximize $Z = 5x_1 + 4x_2$ subject to $0 \leq x_1 \leq 12$, $0 \leq x_2 \leq 9$, $3x_1 + 6x_2 \leq 66$ and $x_1, x_2 \geq 0$				1	6																																										
Q.3	Solve the following.					12																																										
A)	Using the following cost matrix determine optimal job assignment and the associated cost: TABLE <table> <tr> <th>Machinists</th> <th>Job 1</th> <th>Job 2</th> <th>Job 3</th> <th>Job 4</th> <th>Job 5</th> </tr> <tr> <td>A</td> <td>10</td> <td>3</td> <td>3</td> <td>2</td> <td>8</td> </tr> <tr> <td>B</td> <td>9</td> <td>7</td> <td>8</td> <td>2</td> <td>7</td> </tr> <tr> <td>C</td> <td>7</td> <td>5</td> <td>6</td> <td>2</td> <td>4</td> </tr> <tr> <td>D</td> <td>3</td> <td>5</td> <td>8</td> <td>2</td> <td>4</td> </tr> <tr> <td>E</td> <td>9</td> <td>10</td> <td>9</td> <td>6</td> <td>10</td> </tr> </table>				Machinists	Job 1	Job 2	Job 3	Job 4	Job 5	A	10	3	3	2	8	B	9	7	8	2	7	C	7	5	6	2	4	D	3	5	8	2	4	E	9	10	9	6	10	2	6						
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D	3	5	8	2	4																																											
E	9	10	9	6	10																																											
B)	Find the basic feasible solution of the following transportation problem by north-west corner rule. Also find the optimal transportation plan. <table> <tr> <th></th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>Available</th> </tr> <tr> <td>A</td> <td>4</td> <td>3</td> <td>1</td> <td>2</td> <td>6</td> <td>80</td> </tr> <tr> <td>B</td> <td>5</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>60</td> </tr> <tr> <td>C</td> <td>3</td> <td>5</td> <td>6</td> <td>3</td> <td>2</td> <td>40</td> </tr> <tr> <td>D</td> <td>2</td> <td>4</td> <td>4</td> <td>5</td> <td>3</td> <td>20</td> </tr> <tr> <td>Required</td> <td>60</td> <td>60</td> <td>30</td> <td>40</td> <td>10</td> <td>200 Total</td> </tr> </table>					1	2	3	4	5	Available	A	4	3	1	2	6	80	B	5	2	3	4	5	60	C	3	5	6	3	2	40	D	2	4	4	5	3	20	Required	60	60	30	40	10	200 Total	2	6
	1	2	3	4	5	Available																																										
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Q. 4	Solve Any Two of the following.					12																																										
A)	What is mean by inventory? Explain necessity of maintaining inventory.				3	6																																										
B)	The demand for an item is 8000 units per annum and the unit cost is Rs. 1/- the inventory carrying charges of 20% of average inventory cost and ordering cost is Rs. 12.5 per order. calculate optimal order quantity, optimal order time, optimal inventory cost and no. of orders				3	6																																										

C)	A company requires 16000 units of raw material costing of Rs. 2/- per unit. The cost of placing an order is of Rs. 45/- and carrying costs are 10% per year unit of average inventory. Find 1. Economic order quantity. 2. Cycle time 3. Total variable cost of managing inventory	3	6																											
Q.5	Solve Any Two of the following.		12																											
A)	1. Define queuing system, with their assumptions. 2. Discuss the basic terminology of waiting line model.	4	6																											
B)	An auto-rickshaw driver finds from his previous records that the cost per year of running an auto-rickshaw whose purchase price is Rs. 7,000 is as given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>Running Cost(Rs.)</td><td>1100</td><td>1300</td><td>1500</td><td>1900</td><td>2400</td><td>2900</td><td>3500</td><td>4100</td></tr><tr><td>Resale price(Rs.)</td><td>3100</td><td>1600</td><td>850</td><td>475</td><td>300</td><td>300</td><td>300</td><td>300</td></tr></table> At what age is the replacement due?	Year	1	2	3	4	5	6	7	8	Running Cost(Rs.)	1100	1300	1500	1900	2400	2900	3500	4100	Resale price(Rs.)	3100	1600	850	475	300	300	300	300	4	6
Year	1	2	3	4	5	6	7	8																						
Running Cost(Rs.)	1100	1300	1500	1900	2400	2900	3500	4100																						
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C)	A branch of Punjab National Bank has only one typist. Since the typing work varies in length (number of pages to be typed), the typing rate is randomly distributed approximating a Poisson distribution with mean service rate of 8 letters per hour. The letters arrive at a rate of 5 per hour during the entire 8-hour work day. If the typewriter is valued at Rs. 1.50 per hour, determine 1. Equipment utilization. https://www.batuonline.com 2. The per cent time that an arriving letter has to wait. 3. Average system time. 4. Average cost due to waiting on the part of typewriter i.e., it remaining idle.	4	6																											
Q. 6	Solve Any Two of the following.		12																											
A)	Explain Resource allocation and Resource leveling in detail.	6	6																											
B)	Discuss applications of CPM/PERT in project planning and explain the difference between them.	5	6																											
C)	The utility data for a network are given below. Determine the total, free, independent and interfering floats and identify the critical path. <table><tr><td>Activity</td><td>0-1</td><td>1-2</td><td>1-3</td><td>2-4</td><td>2-5</td><td>3-4</td><td>3-6</td><td>4-7</td><td>5-7</td><td>6-7</td></tr><tr><td>Duration</td><td>2</td><td>8</td><td>10</td><td>6</td><td>3</td><td>3</td><td>7</td><td>5</td><td>2</td><td>8</td></tr></table>	Activity	0-1	1-2	1-3	2-4	2-5	3-4	3-6	4-7	5-7	6-7	Duration	2	8	10	6	3	3	7	5	2	8	5	6					
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